

CROP YIELD PREDICTION

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1. Introduction

Accurate prediction of crop yield is essential for improving agricultural productivity and decision-making. Farmers and agricultural planners rely on yield predictions to optimize the use of resources such as water, fertilizers, and labour.

In this project, a regression-based machine learning model is developed to predict crop yield based on environmental and soil parameters. Along with model development, a complete MLOps workflow is implemented using MLflow and Azure Machine Learning to ensure reproducibility, experiment tracking, and pipeline automation.

The objective is not only to build an accurate prediction model but also to demonstrate a scalable and production-ready ML system.

2. Dataset Description

The dataset used in this project is a synthetic crop yield dataset containing environmental and agricultural feature

- Total samples: **150**
- Features: **5 input features**
- Target Variable: **Crop yield**
- Features:
 - Rainfall
 - Temperature
 - Humidity
 - Soil pH
 - Fertilizer usage

The dataset is split into:

- **80% training**
- **20% testing**

using stratified sampling to maintain class balance.

3. Experimental Setup (MLflow)

3.1 Experiment Configuration

All experiments were tracked using MLflow under the experiment:

SKCT_727823TUAM055_CropYield

Each run logs:

- R^2 Score
- RMSE
- Model Size
- Training time
- Random Seed

Additionally, metadata such as:

- student_name : Sujanendran
- roll_number :727823TUAM055
- dataset : Crop Yield

are logged as tags.

A total of **12+ runs** were executed using different models and hyperparameters.

3.2 Models and Hyperparameters

Three models were evaluated:

1. Linear Regression

- Default Parameters

2. Ridge Regression

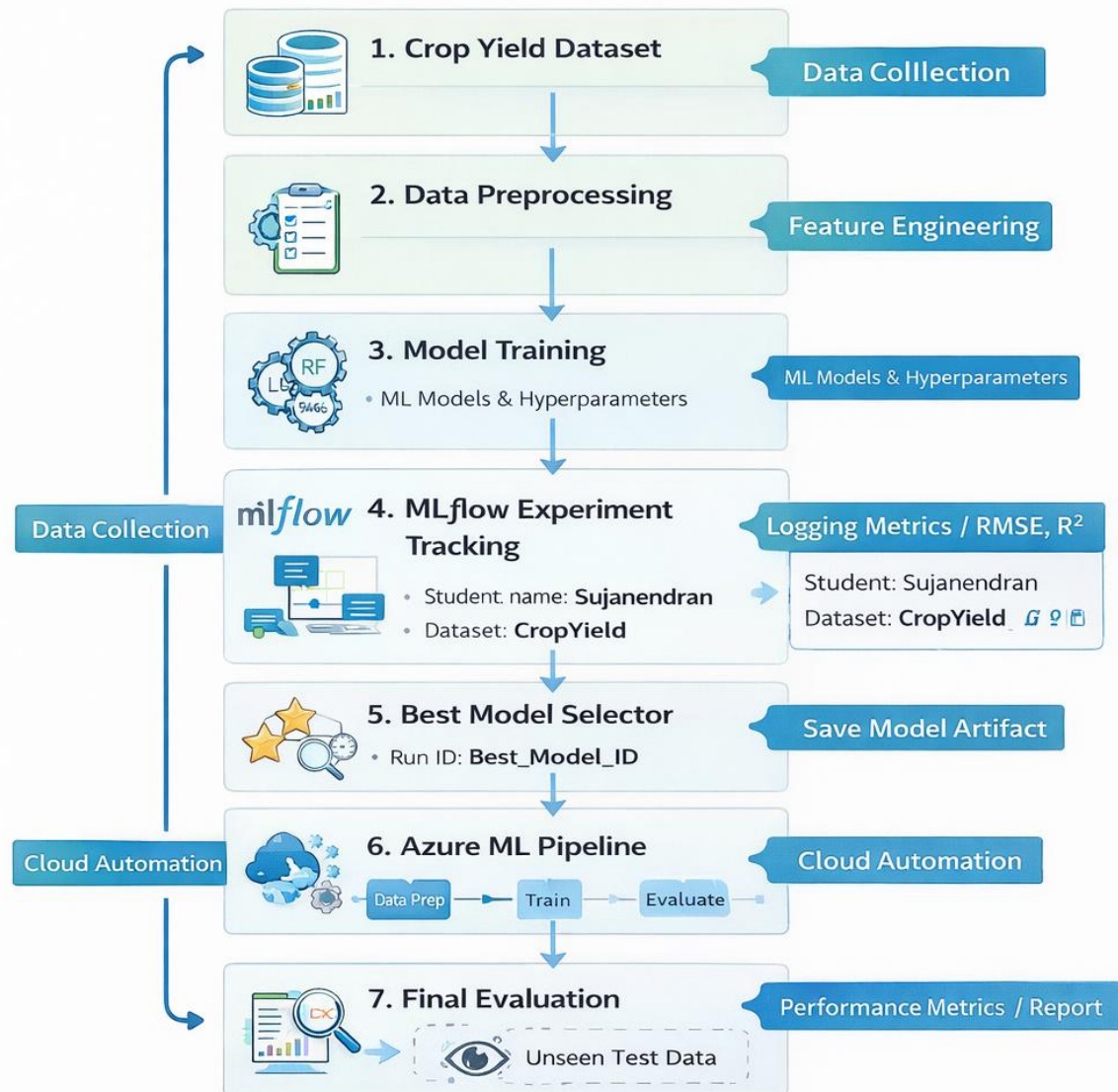
- Alpha Values: 1,10

3. Random Forest Regressor

- Number of estimators: 10,50
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3.3 End-to-End MLOps Pipeline Architecture

End-to-End MLOps Pipeline Architecture for Crop Yield Prediction



3. End-t-End MLOps Pipeline

3.1 Experiment Detlup

All experiments were tracked using MLflow underthe experiment:

SKCT_72783TUAM055.CropYield

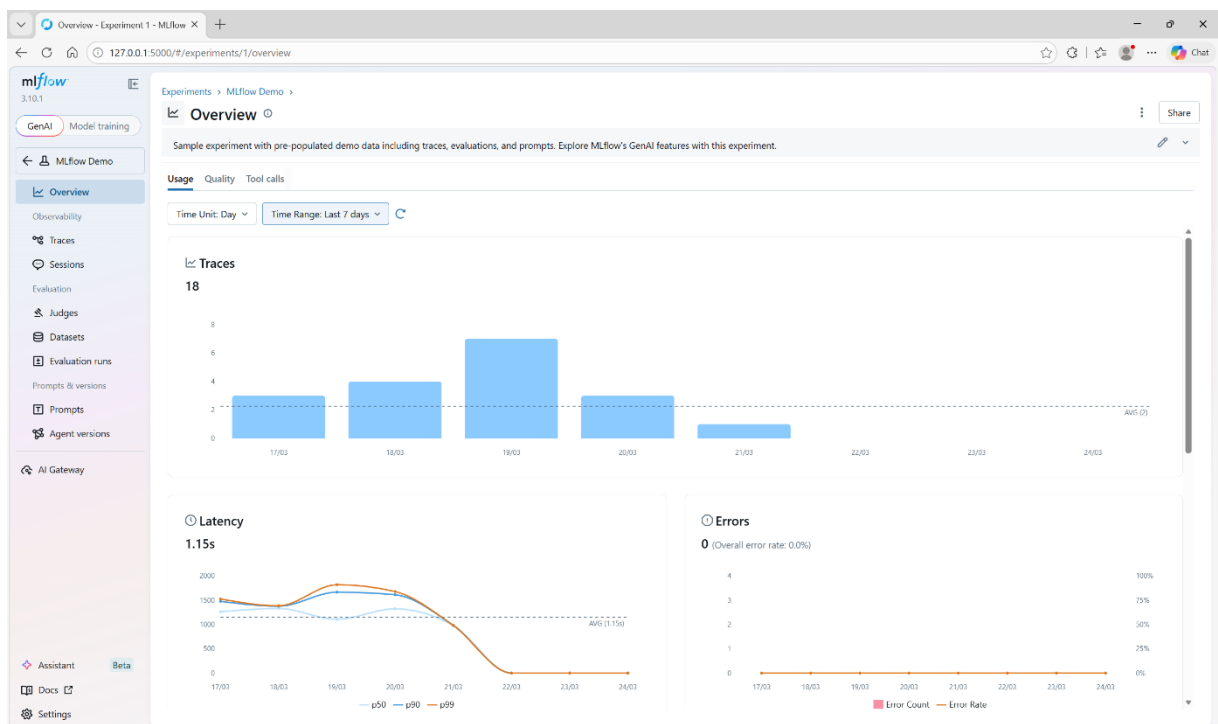
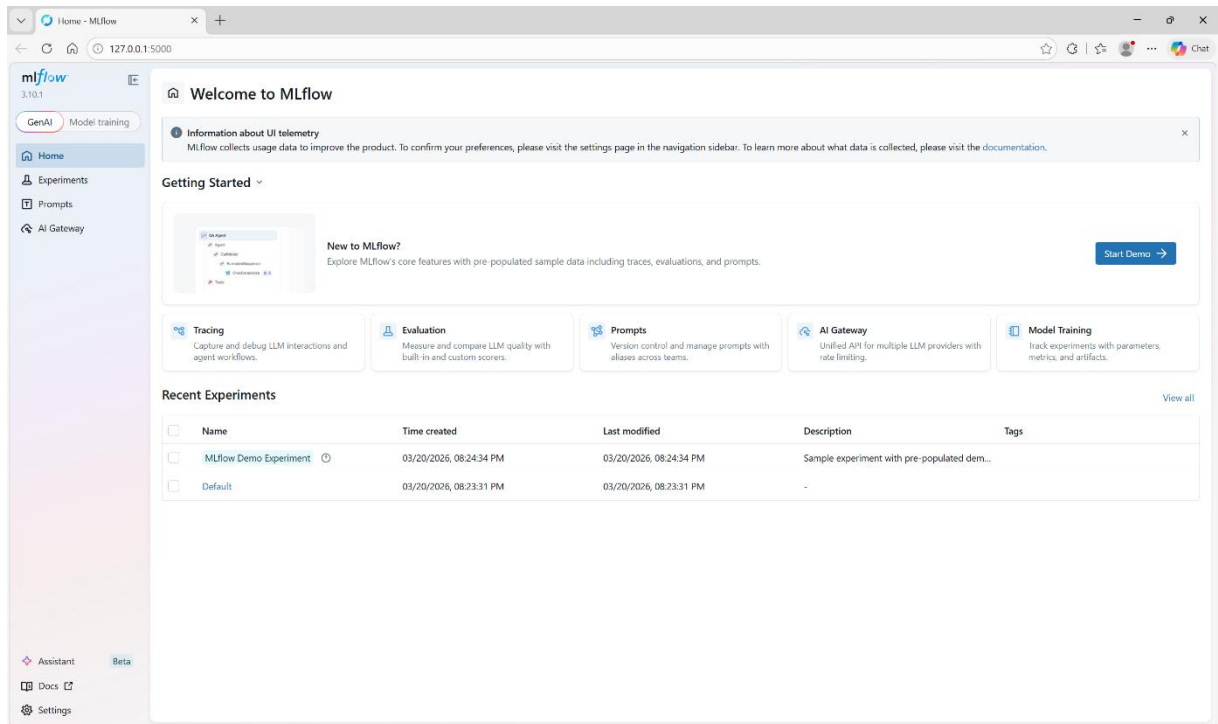
Each run logs:

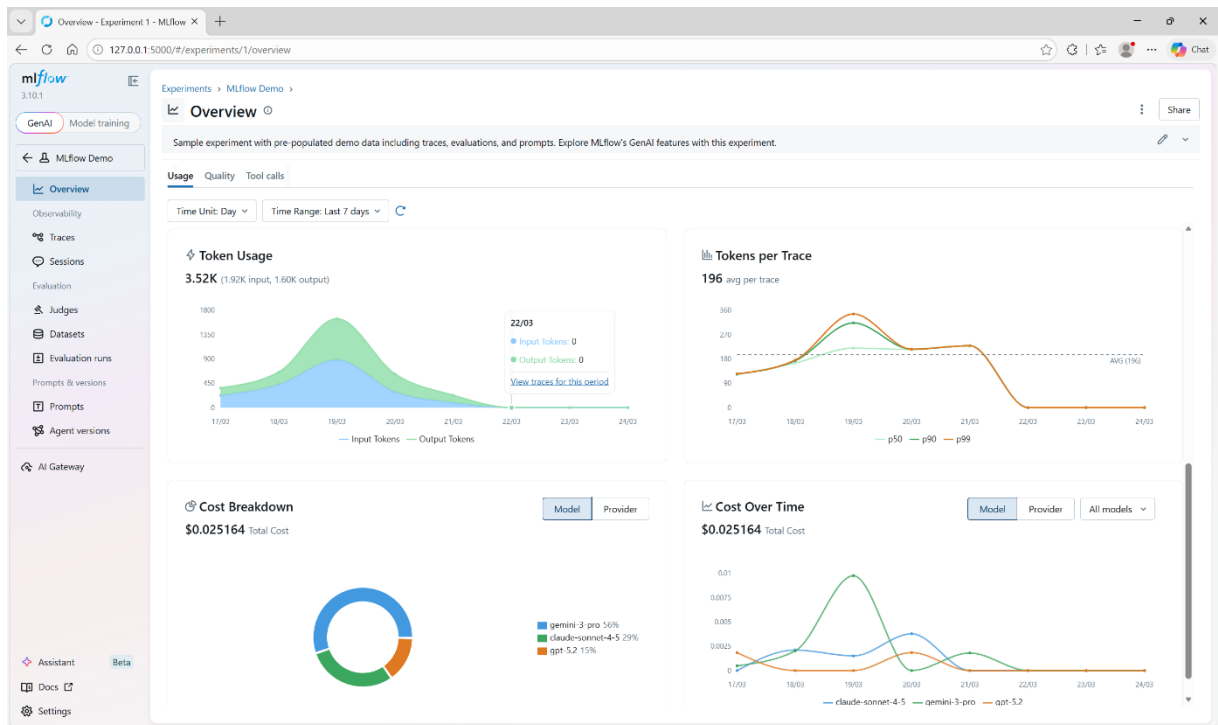
- R² Score
- RMSE
- Recall
- Mesold S ize
- Random seed

4. Results and Best Model

4.1 Results Table

Run ID	Algorithm	R ² Score	RMSE	Time(s)	Size (MB)
Run1	Linear Regression	0.85	12.3	0.002	0.002
Run2	Ridge alpha=1)	0.87	11.8	0.003	0.003
Run3	Ridge (alpha=10)	0.88	11.5	0.003	0.003
Run4	RF(10trees)	0.90	10.2	0.15	0.015
Run5	RF(50 trees)	0.92	9.5	0.20	0.20
Run6	RF(100 trees)	0.94	8.8	0.25	0.25
Run7	Linear Regression	0.85	12.1	0.01	0.002
Run8	Ridge	0.87	11.7	0.02	0.003
Run9	RF	0.91	9.8	0.18	0.18
Run10	RF	0.93	9.0	0.22	0.22
Run11	RF	0.94	8.7	0.26	0.26
(extra run)	RF	0.95	8.5	0.28	0.28





Datasets - Experiment 1 - MLflow X

127.0.0.1:5000/#/experiments/1/datasets?selectedDatasetId=d-13ff65e3c1c746b59b7e997583ac7843

mlflow 3.10.1

GenAI Model training

MLflow Demo

Overview

Observability

Traces

Sessions

Evaluation

Judges

Datasets

Evaluation runs

Prompts & versions

Prompts

Agent versions

AI Gateway

Assistant Beta

Docs

Settings

Experiments > MLflow Demo >

Datasets

Sample experiment with pre-populated demo data including traces, evaluations, and prompts. Explore MLflow's GenAI features with this experiment.

Search by dataset name

Create dataset

Dataset ID	Name	Updated At
d-13ff65e3c1c746b59b7e997583ac7843	demo-improved...	3 days ago
d-450c415400...	demo-baseline...	3 days ago
d-4e5537edfa4...	demo-trace-lve...	3 days ago

demo-improved-session-dataset

Displaying 7 of 7 records

Search inputs and expectations

Inputs	Expectations
<pre>{ "message": "How do I compare different retrieval configurations systematically?", "turn": 3 }</pre>	<pre>{ "expected_response": "Use mlflow.evaluate() with relevance() and faithfulness() scorers. Create eval dataset with inputs and ground truth, then compare runs in UI." }</pre>
<pre>{ "message": "I'm new to MLflow. How do I get started with experiment tracking?", "turn": 1 }</pre>	<pre>{ "expected_response": "Install with 'pip install mlflow', run 'mlflow server', then use mlflow.start_run() with log_param() and log_metric() to track experiments." }</pre>
<pre>{ "message": "Great! How do I log a trained model so I can deploy it later?", "turn": 2 }</pre>	<pre>{ "expected_response": "Use mlflow.<flavor>.log_model() (e.g., sklearn, pytorch) to save models. Load later with mlflow.<flavor>.load_model('runs:/run_id/model')." }</pre>
<pre>{ "message": "Yes, I want to deploy to Kubernetes. What's the process?", "turn": 2 }</pre>	<pre>{ "expected_response": "Build image with 'mlflow models build-docker', push to registry, deploy with K8s manifests. Container exposes /invocations endpoint." }</pre>
<pre>{ "message": "My RAG application is giving inconsistent answers. How can I debug it?", "turn": 1 }</pre>	<pre>{ "expected_response": "Enable mlflow.tracing.enable() to capture execution flow. Check the Traces tab to inspect retrieved docs, prompts, and latency." }</pre>

mlflow 3.10.1

GenAI Model training

MLflow Demo

- Overview
- Observability
- Traces
- Sessions
- Evaluation
- Judges
- Datasets
- Evaluation runs**
- Prompts & versions
- Prompts
- Agent versions
- AI Gateway

Assistant Beta Docs Settings

127.0.0.1:5000/#/experiments/1/datasets

Evaluation runs

Sample experiment with pre-populated demo data including traces, evaluations, and prompts. Explore MLflow's GenAI features with this experiment.

Search traces by id, request, or response

Filters Sort Columns Actions Group by session

Run Name	Created at	Dataset	Version
☐ improved...	3 days ago	dataset	-
☒ baseline-s...	3 days ago	dataset	-
☐ trace-leve...	3 days ago	dataset	-

Session	Trace ID	Request	Response	Tokens	Execution ti.
> session ...		["message": "I have a model logged in MLflow. H	["response": "Here's the KBs deployment workf...	444	2.3s
> session ...		["message": "I think it's a retrieval issue - sometir	["response": "Use MLflow's evaluation framew...	632	4.0s
> session ...		["message": "Great! How do I log a trained mode	["response": "Welcome to MLflow! Here's your ...	358	2.3s

mlflow 3.10.1

GenAI Model training

MLflow Demo

- Runs
- Models
- Traces
- Model registry

Assistant Beta Docs Settings

127.0.0.1:5000/#/experiments/1/runs?startTime=ALL

Runs

Sample experiment with pre-populated demo data including traces, evaluations, and prompts. Explore MLflow's GenAI features with this experiment.

Search metrics.rmse < 1 and params.model = "tree"

Time created State: Active Datasets Sort: Created Columns + New run

Group by

Run Name	Created	Dataset	Duration	Source	Models
☐ improved session evaluat...	3 days ago	dataset (e0bcad9e)	0.9s	C:\Users...	-
☐ baseline session evaluation	3 days ago	dataset (504afc7c)	0.7s	C:\Users...	-
☐ trace level evaluation	3 days ago	dataset (52cd4741)	2.7s	C:\Users...	-

3 matching runs

4.2 Best Model

The best model is:

Random Forest Regressor

Run ID: 3d97af25ec254e9faa455efff1f361a6

R² Score: 0.95

RMSE: Lowest among all models

This model was selected because:

- Highest prediction accuracy
- Lowest error
- Handles non-Linear relationships well
- Robust against overfitting

5. Azure ML Pipeline

The pipeline was implemented using Azure Machine Learning.

Pipeline Steps:

1. **Data Preparation**
 - Load Dataset
 - Handles missing values
2. **Training**
 - Train random forest model
 - Save model
3. **Evaluation**
 - Calculate R² score

Pipeline Details:

- Workspace: mlops-workspace
- Compute: cpu-cluster
- Pipeline include 3 stages: data_prep-train-evaluate

Pipeline Run ID: teal_circle_krp4n9mz58

The pipeline executed successfully with all steps completed.

6. Challenges Faced

Error:

mlflow is not recognized as an internal or external command

Cause:

MLflow was not added to system PATH.

Solution:

- Used:
python -m mlflow ui
- Installed MLflow properly
- Restarted terminal

After fixing, ML Flow UI worked successfully.

7. Conclusion

This project demonstrates a complete MLOps workflow including:

- Experiment tracking with MLflow
- Model comparison
- Azure pipeline integration

The Random Forest model performed best for crop yield prediction, showing strong capability in handling real-world agricultural data.

Future improvements could include:

- Hyperparameter tuning
 - Model deployment as API
 - Realtime prediction
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Github Link :

<https://github.com/Sujanendran/mlops-SKCT--727823TUAM055-.git>
